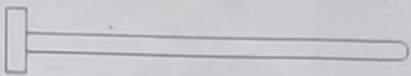
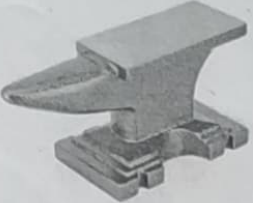
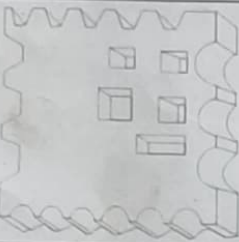
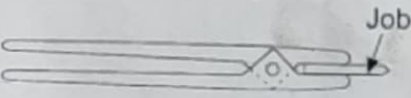
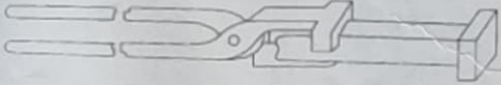
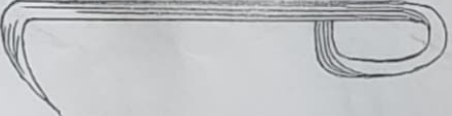
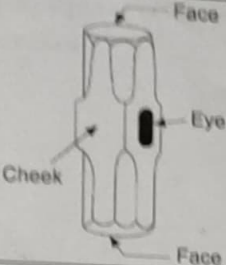
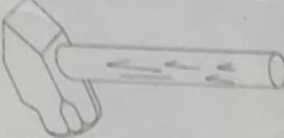
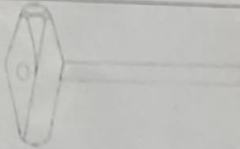
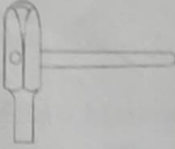


Manufacturing Practice Workshop I

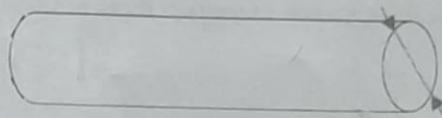
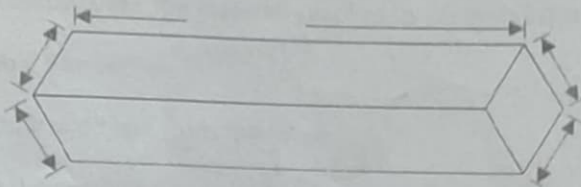
Blacksmithy Shop Manual

Aim 1: To identify the equipments and tools of Smithy Shop

Sl. No	Tools	Name of the tool	Type of Tool (Holding tool/ Striking tool/Hand Tools)
1			
2			
3			
4			
5			
6			
7			
8			

9			
10			
11			
12			
13			
14			
15			

Aim 2: To create a square rod from a given round rod by performing hand forging operations and viceversa.



Tools Required:

- Smith's forge
- Anvil
- Ball-peen hammers
- Flatters
- Swage block
- Half-round tongs
- Pickup tongs
- Cold chisel

Sequence of Operations:

1. Preparation of Raw Material:

- Select a mild steel rod with a 100 mm diameter from the stock.
- Cut the rod to a length of 75 mm.

2. Heating the Material:

- Secure the round rod using round tongs.

- Heat the rod in a blacksmith's forge until it reaches a red cherry color, indicating the correct forging temperature.

3. Reaching Recrystallization Temperature:

- Continue heating until the material reaches its recrystallization temperature.

4. Forging the Square Shape:

- Remove the heated rod from the forge.
- Place it on the anvil and use a sledgehammer to hammer the rod evenly, forming square edges on all sides.

5. Hammering Process:

- Perform hammering on the anvil surface to ensure even shaping.

6. Final Shaping:

- If any bends occur, reheat the rod as needed and continue hammering to achieve the desired square shape.

7. Cooling and Inspection:

- After forging, quench the rod to cool it.
- Measure and check the dimensions of the square rod to ensure accuracy.

Result:

The square rod was successfully created from the given round rod.

Precautions:

1. Handle the workpiece carefully during heating and hammering.
2. Ensure the workpiece is held parallel to the face of the anvil while hammering.
3. Wear steel-toed shoes to protect your feet.
4. Use a face shield to protect your face from hot metal splashes.
5. Select the correct size and type of tongs to securely hold the workpiece.